[17] The Catapult Mechanism at Large Learning Rates

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The Large Learning Rate Phase of Deep Learning: The Catapult Mechanism — RQ2, RQ4

Problem: Why can very large learning rates train deep networks successfully, sometimes better than small learning rates, even though the first steps may increase loss? **Method:** Derive dynamics in a solvable class of wide networks and test predictions in neural-network experiments. **Result:** A large-but-stable learning-rate regime can cause an initial loss *catapult*, accompanied by a decrease in curvature, followed by successful training in flatter regions. This is a model-dependent mechanism supported by experiments, not a universal guarantee.

1. The standard intuition: too large a step should fail

A local quadratic model with curvature $\lambda > 0$ gives

$$e_{t+1} = (1 - \eta\lambda)e_t.$$

A fixed quadratic becomes unstable for $\eta > 2/\lambda$. If the network behaved like its linearization at initialization forever, then the *initial* curvature λ_0 would seem to determine which learning rates can work. But neural networks are not fixed quadratics: their tangent kernel and curvature evolve.

2. What actually changes during training?

For squared error on training outputs $f_t \in \mathbb{R}^n$, define residuals $r_t = f_t - y$ and the parameter Jacobian $J_t = \partial f_t/\partial\theta$. The empirical tangent-kernel matrix is

$$K_t = J_t J_t^\top.$$

The usual linearized approximation holds J_t fixed, giving *approximately*

$$r_{t+1} \approx (I - \eta K_0)r_t.$$

If K_t evolves substantially, this fixed-kernel approximation ceases to describe the trajectory. The model in the paper uses this freedom: a large step may initially amplify the output error, but the resulting nonlinear changes can reduce the effective curvature so subsequent steps stabilize.

3. The three qualitative learning-rate phases

The authors distinguish a *lazy/small-rate regime*, a *catapult regime*, and a regime with rates so large that stable training fails. In a tractable model, important boundaries are expressed in terms of its initial kernel eigenvalue λ_0, with catapult behavior starting near $2/\lambda_0$; the upper edge of the stable interval *depends on the model*. Do not assume every architecture has the same upper threshold.

Characteristic trajectory:

$$\underbrace{L_0 \to L_{\text{spike}}}_{\text{transient loss increase}} \quad \to \quad \underbrace{K_t \text{ decreases}}_{\text{curvature changes}} \quad \to \quad \underbrace{L_t \text{ falls}}_{\text{stable training}} .$$

The paper reports that larger learning rates can select a flatter solution and sometimes achieve better test performance. The generalized statement "large learning rates always generalize better" is *not* supported.

4. First-principles teaching analogy (not the exact paper model)

Take a scalar toy loss $L(w) = \frac{1}{2}(1 - w^2)^2$. The curvature

$$L''(w) = -2 + 6w^2$$

changes when w changes. Thus a step size incompatible with the *initial* curvature can eventually enter a region with different curvature. This toy merely explains how a moving objective curvature differs from a fixed quadratic; it is not the paper's derived catapult system.

5. Limitations and relation to other work

The exact analysis uses a specially solvable neural model and limiting/scaling assumptions; real-architecture evidence is experimental. The phenomenon also differs from Cohen et al.'s edge of stability: a *catapult* emphasizes an early transient loss spike and curvature reduction, whereas *EoS* describes sustained training near $2/\eta$. Both show why fixed-kernel, tiny-step reasoning misses important dynamics. **Our takeaway.** A learning rate is not just how far GD moves: a sufficiently large step can reshape the part of the loss landscape encountered during optimization. **Source:** https://arxiv.org/pdf/2003.02218 (Sections 2–5).